

Variables in Statistics

Tutorial 3

1. What is a Variable?

Definition: A variable is a property that can take on multiple different values.

Hello students! Today we are learning one of the most important basics of Statistics: **Variables**. Think of a variable as an empty container or a blank space in a form. The label on the container is the "Variable", and the things you put inside are the "Values". Because the values can change or vary, we call it a *variable*!

Real-Life Example:

Example 1 (Age): Let's say we have a variable called **Age**. The value of Age is not fixed. It can take multiple values like 12 years, 30 years, 40 years, or even 100 years!

Example 2 (Height): The variable **Height** can be 102 cm, 172.5 cm, or 180 cm.

Example 3 (Weight): The variable **Weight** can take multiple values like 72 kg, 72.5 kg, or 100 kg.

Important Note & Extra Info:

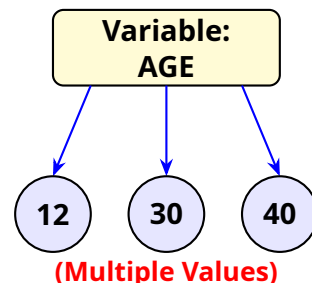
Teacher's Tip - Singular vs. Plural:

A variable is always spoken in the **Singular** form. We say "Age" (Variable), not "Ages". The word "Ages" refers to a collection of data points (e.g., 40, 50, 60), but the property itself is just "Age". This is a small but very important statistical rule!

Practice Question:

Q: In a class, the teacher notes down the "Eye Color" of students (Black, Brown, Blue). Is "Eye Color" a variable?

A: Yes! It is a property that takes on multiple different values (Black, Brown, Blue).



Topic Summary: What is a Variable?

A **Variable** is a singular property (like Height, Weight, or Age) that can hold many different changing values.

2. Types of Variables

In Statistics, variables are divided into two main families based on whether they use **Numbers** (Quantity) or **Labels/Categories** (Quality). Let's explore them in detail!

Type A: Quantitative Variables (Numbers)

Definition: Variables that measure a numerical quantity or amount.

If a variable answers the question "How much?" or "How many?", it is a Quantitative Variable. We can further divide this into two subtypes:

1. Discrete Variables:

- These are variables that can only take **whole numbers** (like 1, 2, 3).
- They cannot have decimals or fractions. You count them!

2. Continuous Variables:

- These variables can take **any infinite value**, including decimals.
- You usually measure them using an instrument (like a scale or ruler).

Real-Life Example:

Discrete Example: Number of bank accounts you have. You can have 1, 2, or 3 accounts. You cannot have 2.5 bank accounts! Another example is the number of children in a family (1 or 2, never 3.5).

Continuous Example: Your Height. You might be exactly 175 cm tall, or you could be 175.25 cm tall! It can take infinite decimal possibilities.

Practice Question:

Q: Is "Number of cars on a road" Discrete or Continuous?

A: Discrete! You count cars in whole numbers (10 cars, not 10.5 cars).

Discrete (Counting)

1 2 3

No decimals!

Continuous (Measuring)

1.5 cm 2.8 cm

Infinite decimal points!

Type B: Qualitative or Categorical Variables

Definition: Variables that classify data into groups based on some properties, not numbers.

These variables do not use numerical math! Instead, they sort things into different categories or labels based on specific characteristics. That is why they are also called "Categorical Variables".

Real-Life Example:

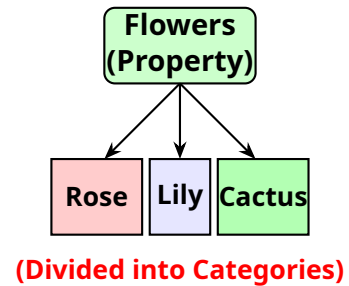
Example 1 (Gender): We can classify people into "Male" and "Female". These are categories based on biological properties, not numbers.

Example 2 (Types of Flowers): If we have a garden, we can group flowers by their type: "Rose", "Lily", or "Cactus". We are dividing them based on their physical properties!

Practice Question:

Q: A teacher records the "Blood Group" (A, B, AB, O) of all students. Is Blood Group Qualitative or Quantitative?

A: Qualitative (Categorical)! Because A, B, AB, and O are category labels, not math numbers.



Overall Revision Summary

- **Variable:** A singular property (like Age, Height) that can take on multiple different values.
- **Quantitative Variables:** Deal with numbers and math.
 - **Discrete:** Whole numbers only (e.g., Number of cars = 2). You *count* them.
 - **Continuous:** Can have infinite decimal values (e.g., Weight = 72.5 kg). You *measure* them.
- **Qualitative / Categorical Variables:** Deal with labels and categories instead of numbers.
 - They divide items based on properties (e.g., Gender = Male/Female, Flower = Rose/Lily).
- **Bonus Note:** Krish Naik Sir has launched a free website (krishnaik.in) to download all these materials and blogs for free!

Keep practicing! Knowing the types of variables is the foundation of Machine Learning and Data Science!